The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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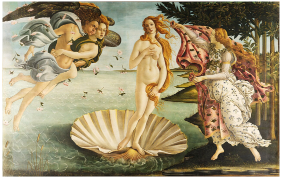
The rooms everyone queues for

Almost every one of the ∼5,000 daily visitors comes for the same handful of paintings, crammed into just three rooms of the ninety-eight.

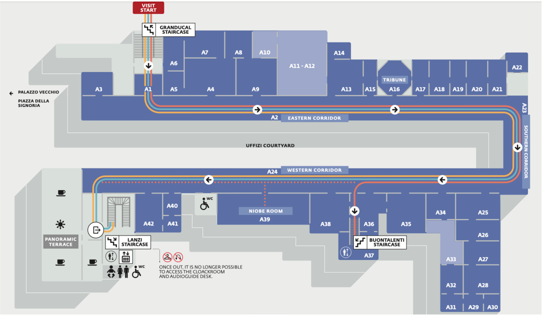


Botticelli's *Primavera* and *Birth of Venus* (room A11/A12); Leonardo's *Annunciation* and *Adoration of the Magi* (A35); Raphael's *Madonna del Cardellino* with Michelangelo's *Doni Tondo* (A38).

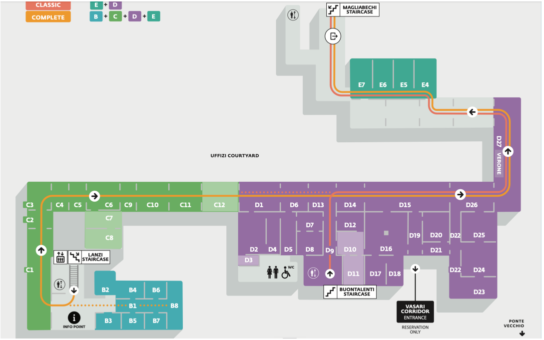
Too many people for a few rooms

The building is large: two floors and ninety-eight rooms. The problem is not the headcount but the *concentration*: demand collapses onto the three masterpiece rooms upstairs and Caravaggio downstairs, so those rooms run far over capacity while the rest sit calm. The crowd is itself mixed, *60%* Instagram tourists, *30%* normal tourists and *10%* art lovers.

Second floor (the three gated masterpieces)



First floor (Caravaggio, free-flow)



The real Gallerie degli Uffizi plan: the famous rooms cluster off one corridor on each floor.

Two streams, one project

We split the project into two parts that stay separate until the very end.

Stream 1: crowd simulator.

Run the whole 60/30/10 population through a full day, minute by minute.

- Compare the museum *as it is* against a Pigovian redesign (booked slots, pricing, hours).
- The *director's gate*: adopt it only if welfare rises and revenue does not fall.
- It passes: welfare $+31\%$, revenue preserved.

Stream 2: RL on top (our focus).

Take that redesigned museum as fixed and ask the sharper question.

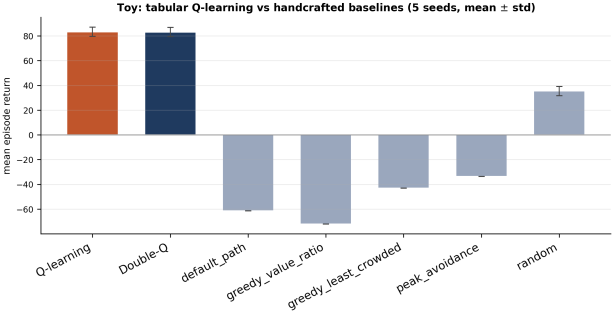
- Not the average visitor: the *optimal* one.
- When to book, which masterpieces and how to pace the day.
- All under genuine uncertainty about the crowd.

The gate passes first; only then do we study the optimal visit.

Foundations: a tabular proof of concept

We first prove the idea on a *12-room toy*, small enough to keep an exact table of values for every situation. Crowds rise and fall in fixed waves; the agent is never told the schedule. Trained by trial and error, tabular *Q-learning* discovers *temporal arbitrage*: enter a masterpiece in the lull before or after its crowd peak, not when a guidebook would send everyone.

Honest result (5 seeds): Q and *Double-Q* come out *tied* (83.3 vs 83.1). On a deterministic toy there is no overestimation bias for Double-Q to remove, so it neither helps nor hurts, exactly as the theory predicts.



Learning beats every hand-written rule by a wide margin; the structured rules even go negative because they reliably mistime the crowd.

Timing, learned from samples alone, beats every hand-written rule.

Scaling up: deep RL and action masking

The real museum destroys the table: ninety-eight rooms, about a hundred possible moves at each step and a state with several hundred numbers. We replace the table with a neural network and train it with *MaskablePPO*, keeping unmasked PPO and DQN as controls so any gap is about the method, not luck.

Action masking: hand the agent the rulebook

The museum is a graph, so from any room only a few moves are legal, yet the network has $\sim$100 outputs. We set the score of every illegal move to $-\infty$ before the softmax, so it gets zero probability. No gradient is spent learning which moves exist; all of it goes into the decision that matters. This is mathematically a state-dependent action set and it is the single biggest architectural choice in the project.

It turns out to be decisive exactly where the decision is hardest.

The decisive reframe: a booking problem

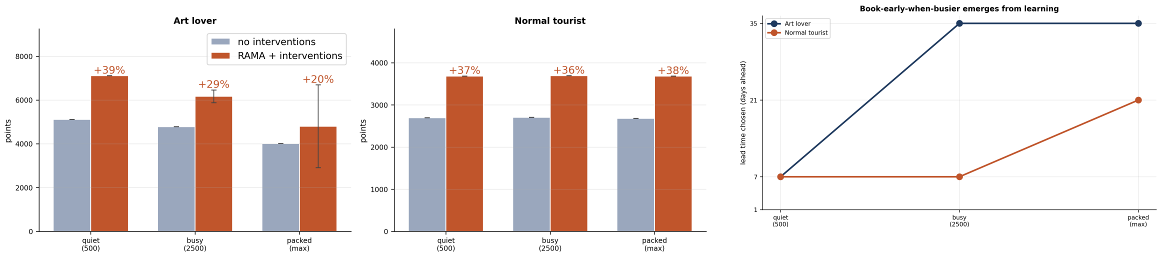
We spent months on a *navigation* agent before admitting the obvious: a real visitor has a map and is guided by staff, so optimising the walk is a shortest path dressed up as RL, a trivial, fully observed problem. The genuinely uncertain, consequential decision is the *booking*.

- *Baseline (the museum as-is):* no booking at all, just walk into the crowds and take the masterpieces as you find them.
- *Intervened (RAMA):* the masterpieces are gated by booked time-slots, so the decision is the *lead time* (1, 7, 21, or 35 days ahead), then pacing to reach each booked window.
- A crowd-dependent *fill curve*: on a busy day the good slots sell out early, so booking late forfeits masterpieces entirely. The discount is $\gamma = 0.999$, since the payoff lands at check-in.

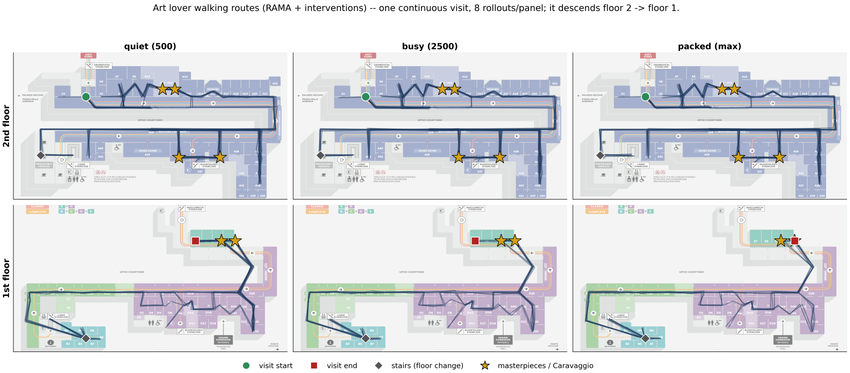
Choosing the right MDP, not tuning the algorithm, was more than half the project.

Result: book early when busier and it pays

RAMA booking: intervened vs matched baseline (5 seeds, mean ± std; 3/3 masterpieces secured)

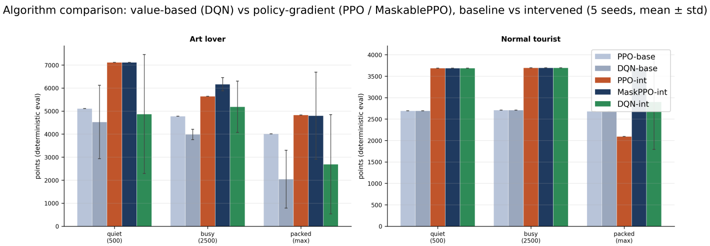


Both profiles beat the matched no-booking baseline at every crowd level (five seeds, mean$\pm$std): +20% to +39% for the art lover, +36% to +38% for the tourist, with all three masterpieces secured. The learned lead time *rises with the crowd* (right): book-early-when-busier is not coded in, it emerges from the reward, because booking late at a packed museum misses the rooms.



The art lover's learned visit, eight rollouts per panel, across the three crowd levels. It is one continuous, sensible route: enter at `A1`, sweep the second-floor masterpieces (gold stars) in their booked windows, descend the staircase to Caravaggio on the first floor and leave before closing. Every step is a one-hop move to a neighbouring room, so there are no teleports.

Which algorithm wins and why masking matters



- *MaskablePPO* is best or tied in every cell and stable across seeds.
- *Masking is decisive exactly at the hardest cell:* at the packed tourist, unmasked PPO *collapses* to 2093, below the no-booking baseline of 2680, while MaskablePPO holds 3685. At easier crowds the two agree; the gap appears only where precision matters.
- *DQN is crowd-fragile and high-variance* (its baseline falls 4527 $\rightarrow$ 2044): taking a max over noisy crowd values overestimates, the very bias the deterministic toy did not have.

The journey was the project: 1, an unlearnable reward

S *Symptom.* The agent never left the museum: it would squat in one room until the day ended. The obvious fix, zeroing all reward if it failed to exit, made the behaviour *worse*, not better.

D *Diagnosis.* An all-or-nothing terminal wipe is *flat* across almost every policy, so the gradient is zero. There is a cliff at the exit and a plateau everywhere else and from the plateau nothing points toward the door.

F *Fix.* Replace the wipe with a *dense* egress signal: a per-step pressure that grows as closing nears and as you get further from an exit, a finite (not infinite) no-exit penalty and a bonus for leaving.

L *Lesson.* Sparse, all-or-nothing rewards are correct but often unlearnable. Shape a slope the optimiser can actually climb.

2: the agent could not see what it had to decide on

S *Symptom.* The rule "stay until satisfied, then move on" never stabilised; the agent over- and under-dwelt rooms seemingly at random, even with a sensible reward.

D *Diagnosis.* The right action depended on *how much value it had already extracted from this room* and that progress was *not in the observation*. It was a hidden variable, so the task was a POMDP, not the MDP we thought we had.

F *Fix.* Add a per-room *appreciation-progress* value to the state, so "have I seen enough of this room?" becomes a function of something the policy can actually read.

L *Lesson.* The state must contain everything the optimal action depends on, or you are not solving the problem you think you are.

3: reward shaping that backfired

S *Symptom.* Well-meant extra terms made things worse: a "boredom" penalty produced *looping* and a harsh crowd penalty produced a *fake art lover* that sprinted past crowded masterpieces.

D *Diagnosis.* Every term you add is a new objective the optimiser will game (reward hacking): circling a loop beat exiting under the boredom penalty and a crowded Botticelli scored worse than skipping it.

F *Fix.* *Minimal* shaping: a *gentle* crowd discount so a crowded masterpiece still beats skipping it, drop the boredom penalty entirely and let the satiation term carry "stop lingering" on its own.

L *Lesson.* Stress-test every reward term by asking what the cheapest way to collect it is and whether that is what you actually want.

4: the do-nothing trap and the trick that cracked book-early

S *Symptom.* Given a free “decline” button, *both* profiles learned to decline all three masterpieces. And even without it, book-early was only *half* learned, with a flat spread of lead times.

D *Diagnosis.* Declining is immediately safe while booking only pays off much later, a do-nothing local optimum; and the booking reward lands hundreds of steps away, so its gradient on the booking action is faint (long-horizon credit assignment).

F *Fix.* Treat “always books” as a *type trait* that rules the escape hatch out and add an *immediate off-pace penalty* at booking time, a local proxy that correlates with the distant cost.

L *Lesson.* A small immediate signal that tracks a far-off reward is what makes a long-horizon lesson learnable. This was the single most decisive trick.

What five seeds revealed (and why we ran them)

A single training run can be lucky or unlucky, so a single number can mislead. We retrained every cell under *five* training seeds and report the mean and the spread across them.

- *The tourist is rock-stable*: essentially zero variance at every crowd level.
- *The art lover degrades as the museum fills*: the gain falls from +39% to +29% to +20%.
- *The honest catch:* at *maximum crowd* the art lover is *bimodal*, four of the five seeds reach ∼+41% but one collapses below baseline, so the cell averages +20% with a wide error bar. We report the spread, not the lucky seed.

Reporting variance honestly is part of the result, not a footnote.

What worked best and worst

Worked best

- *Action masking*, the single biggest lever.
- *Reframing* navigation into booking: the right MDP.
- *An immediate proxy* for the delayed reward.
- *Exposing appreciation progress*: Markov again.
- *Dense egress shaping*: a learnable way out.

Worked worst

- *All-or-nothing exit wipe*: zero gradient.
- *A free decline button*: a do-nothing trap.
- *A harsh crowd penalty*: a fake art lover.
- *A boredom penalty*: a perverse looping incentive.
- *Greedy evaluation* of a stochastic policy.

The failures and their diagnoses, are the project; the numbers are where they stopped.

Conclusion and what is debatable

- *The result:* given the director-safe redesign, the optimal art lover and tourist *book early, earlier when busier*, secure all three masterpieces and beat the no-booking baseline by +20% to +39% (art) and +36% to +38% (tourist).

- *The journey was the project:* a chain of instructive failures, each a named RL lesson, unlearnable reward, partial observability, reward hacking, the do-nothing trap, credit assignment.

- *Honestly debatable:* the 60/30/10 split, the dwell and crowd-density calibration and the room positioning are modelling choices (the floor-plan maps let you judge them); the fill curve is calibrated rather than fitted to booking data; and the as-is museum abstracts away the real entry choice (pre-book and skip the queue, or pay less and queue at the door).

Thank you.

Questions?

Code, figures and a plain-language guide to every algorithm:

`github.com/mocl20/Uffizi`